

Ashritha Gugire

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SUMMARY

Python Developer with 4+ years of experience in AI/ML model development, research, and agent-based intelligent systems. Proficient in building, training, and deploying ML models with Python using frameworks like TensorFlow, PyTorch, and Scikit-learn. Experienced in developing autonomous behavior models using graph-based and Agentic AI techniques for real-time anomaly detection and decision-making systems. Skilled in MLOps, data pipelines, and scalable AI system architecture on AWS and communicating effectively to both technical and non-technical audiences.

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, SciPy, Matplotlib, Scikit-learn, TensorFlow, PyTorch, PySpark, NetworkX).

Cloud Databases: AWS S3, Sagemaker, EC2, GCP, Docker, MySQL, SQL Server, PostgreSQL

ML Techniques: Multi-agent architectures, Computer vision, NLP models, Pattern recognition, Hypothesis Testing, Statistical analysis

Other Skills: Data Integration, Text Analysis, Git, A/B testing, Training, CI/CD, inference pipelines, Model deployment

Soft Skills: Collaboration, Presentation Skills, Problem-solving, Critical Thinking and Continuous Learning

PROFESSIONAL EXPERIENCE

Deloitte | Rosslyn, VA | Jan 2024 - Present

Python Developer – Machine Learning R&D

- Developed an **agent based predictive anomaly detection system using Python**, NetworkX and ML models on over 60,000 records, to monitor agents behavior analytics to solve business problems.
- Designed **advanced ensemble framework using XGBoost, Random Forest and AdaBoost** to predict and validate user behaviors over individual models and **enhanced prediction accuracy by 18%**
- Researched and **implemented 15+ knowledge graphs and Agentic AI techniques** to model autonomous agent behavior and temporal dependencies and mobility networks in urban mobility operations.
- Created **advanced statistical models** using time series decomposition, hypothesis testing, PCA, and clustering to interpret anomaly trends **to understand decision-making behavior of agents over time**.
- Conducted **research on advanced machine learning methods to reduce false positives** in data anomalies, translating business requirements into technical solutions that resolved critical pain points
- Delivered **MLOps solutions using MLflow on AWS EC2** for automated deployment and monitoring, providing real-time anomaly streaming tracking to support data-driven decision-making

Infosys | India | July 2020 – Dec 2022

Python Programming Analyst

- Facilitated collaboration between business users and technical team to optimize supply chain operations, streamline database performance, reducing error rates and improving data accuracy by 45% within agile teams.
- Built **sales forecasting and demand planning models using ARIMA time series analysis**, increasing revenue by 20% product classification.
- Engineered ETL pipelines using **SQL queries and python scripts to preprocess large datasets from AWS S3**, improving data retrieval efficiency and utilizing **JIRA and Git** for version control and collaborative project management.
- Designed and executed A/B tests to evaluate feature performance and user engagement, analyzing results with statistical methods that contributed to a 15% increase in customer retention.
- Managed daily business-as-usual activities, including **database maintenance, dashboard troubleshooting, and cross-team collaboration** to ensure smooth workflow.

KEY PROJECTS/ RESEARCH EXPERIENCE

Autonomous Customer Support Agent (Agentic AI, RAG, Python, LangChain, FAISS, OpenAI)

- Designed and built an agentic AI system with autonomous decision-making that processed through RAG pipeline using FAISS vector database, OpenAI embeddings, and LangChain Orchestration, with automatic fallback mechanisms between GPT-3.5-turbo and Hugging Face models achieving sub-2 second response times with 95% query resolution rate and zero-touch automation for document processing across multiple formats.

Multimodal Sentiment Analysis System (Python, PyTorch, Transformers, CLIP)

- Engineered multimodal AI pipeline with autonomous model orchestration** that seamlessly integrates RoBERTa text sentiment analysis, OpenAI CLIP image-text understanding, and DistilRoBERTa emotion detection through intelligent preprocessing workflows, achieving 94.2% text accuracy and 87.6% image accuracy with sub-2 second response times across 9 emotion categories and real-time color/brightness analysis for visual sentiment enhancement.
- Trained intelligent cross-modal validation system with adaptive reasoning capabilities** that automatically detects sentiment disagreements between text and visual content resulting in autonomous disagreement flagging with 89% accuracy for detecting mismatched emotional content and scalable architecture supporting concurrent multi-user sessions with persistent session management.

EDUCATION

Masters in Data Analytics and Engineering | George Mason University | Dec 2024

Bachelor's in computer science and engineering | Vignana Bharathi University | May 2021